Title	**Mathematics for Economics**	Number	AIL7380
Department	SAIDE	L-T-P [C]	3-0-0[3]
Offered for	B. Tech./P.G./Ph.D.	Type	Elective
Prerequisite	None		

Objectives

To introduce doctoral students to the basic mathematics and optimization techniques used throughout microeconomic theory. These mathematical tools will be illustrated from a theoretical microeconomics point of view with corresponding applications

Learning Outcomes

1. Students will be familiarized with the use of real analysis, linear algebra and static optimization techniques used in microeconomic theory so that they can use and build on them in their dissertation.
2. Students will be able to use these tools to analyze theoretical microeconomic problems.

Course Content

Module 1 [Real Analysis (14)]: Real and complex number systems; Basics of topology; Numerical Sequences and Series; Continuity; Differentiation.

Module 2 [Linear Algebra (14)]: Linear algebra and normed vector space; Orthogonal sets; Projection into subspaces; separation of convex sets; orthogonal complements; Spectral theorem; Singular Value Decomposition Theorem; Applications.

Module 3 [Optimization (14)]: Existence Supremum (Maximum) and Infremum (Minimum) of a real valued Function; Constrained optimization: Lagrange's Method; Extension and generalization; Shadow Price; Maximum Value Function; Convex set and their separation as an optimization tool; Concave programming and second order conditions.

Reference Books:

1. Analysis and Linear Algebra: The Singular Value Decomposition and Applications; James Bisgard
2. Finite-dimensional vector spaces; P.R. Halmos
3. Mathematics for Economists; Carl P. Simon and Lawrence E. Blume
4. Fundamental Methods of Mathematical Economics; Alpha. C. Chiang
5. Principles of Mathematical Analysis; Walter Rudin
6. Introduction to Real Analysis; Robert G. Bartle and Donald R. Sherbert
7. Optimization in Economic Theory: Avinash. K. Dixit
8. A First Course in Optimization Theory: Rangarajan K. Sundaram

Title	Indian Economic Development and Policy	Number	LAL7490
Department	School of Liberal Arts	L-T-P [C]	3-0-0
Offered for	UG, PG and PhD	Type	Elective
Prerequisite	None		

Objectives

1. Introduce students to current and evolving policy debates in the backdrop of issues pertaining to economic development of India
2. Familiarize students with economic development issues and policy aspects which are important in professions involving nation-building broadly and policy formulation and implementation specifically.

Learning Outcomes

1. Equipped with understanding of issues in Indian economic development and ensuing evolution of policy debates
2. Able to have a broad perspective of the multifaceted nature of economic development issues and incorporate this in their professional pursuits

Course Content

- Brief overview of economic growth and development policies in India since Independence Agriculture: Policies and Performance; Growth; productivity; agrarian structure and technology; capital formation; trade; pricing and procurement.
- Industry: Policies and performance; production trends; small scale industries; public sector; foreign investment, labour regulation
- Services: Policies and performance, trends, patterns, composition, drivers Employment : Occupational structure in the organised and unorganised sectors; open-, under- and disguised- unemployment (rural and urban); employment schemes and their impact
- Growth and Distribution: Trends in poverty, inequality and unemployment, Policies for eradicating poverty and generating employment
- Population and Economic Development: Demographic trends and issues, Education, Health and Malnutrition
- International Trade: Performance and Policies, trends and patterns of imports and exports, FDI, FII

Text Books:

1. U. Kapila (2022). India Economy Since Independence. Academic Foundation
2. U. Kapila (2022). India Economy: Performance and Policies. Academic Foundation

Reference Readings:

1. Basu, K. (2011). India's Foodgrain Policy: An Economic Theory Perspective, Economic and Political Weekly, Vol (46), 5.
2. Krueger, A. O. (Ed.) (2002). Economic policy reforms and the Indian economy. University of Chicago Press.
3. S. Mehrotra (2015). Realising the Demographic Dividend, Policies to achieve Inclusive Growth in India, Cambridge University Press

Online Course Material:

1. https://onlinecourses.nptel.ac.in/noc22_hs69/preview

Title	**Microeconomics 1**	Number	**AIL6010**
Department	AIDE (CMCE)	L-T-P [C]	3-0-0
Offered for	B. Tech./P.G./Ph.D.	Type	
Prerequisite	None		

Objectives

1. Introduce the fundamental principles of individual decision-making.
2. Formalize the decision-making behavior for consumers as well as producers.
3. Set up the general equilibrium model and price determination.

Learning Outcomes:

1. Understanding of axiomatic approach of economic decision-making.
2. Modeling economic decision-making in a tractable form and analyze the comparative-statics.
3. Understanding of basic market mechanisms and fundamentals behind price determination.

Course Content:

Preferences: axiomatic approach and examples. (4 lectures)

Utility theory: representation theorem and examples. (4 lectures)

Cardinal vs ordinal approaches. (3 lectures)

Utility maximisation problem. (4 lectures)

Demand analysis. (2 lectures)

Expenditure minimisation problem as dual and Hicksian demand. (2 lectures)

Integrability and recoverability (2 lectures)

Weak axiom of revealed preference (2 lectures)

Producer theory: profit maximisation and cost minimisation. (3 lectures)

Welfare theorems (3 lectures)

Market structures: Perfectly competitive market , Monopoly and discrimination (4 lectures)

Strategic aspects in market structure: oligopoly (3 lectures)

General equilibrium. (6 lectures)

Reference Books:

1. A Short Course in Intermediate Microeconomics with Calculus. Feldman, A. & Serrano, R. , CUP, 2018.
2. Microeconomic theory. Mas-colell, A., Whinston, M.D., Green, J. R., Oxford University Press, 1995.
3. A Course in Microeconomic Theory. David M. Kreps, Princeton University Press, 1990.
4. Advanced Microeconomic Theory. G. A. Jehle, P. J. Reny, Prentice Hall, 1991.

Title	**Theoretical Foundations of Econometrics**	Number	AIL7320
Department/School	AIDE (CMCE)	L-T-P [C]	3-1-0 [3]
Offered for	B. Tech., Post-grad, Ph.D.	Type	Elective
Prerequisite	Statistics		

Objectives:

1. To introduce students to fundamental concepts in econometric theory.
2. To provide knowledge of the theoretical underpinnings of applied econometric methods.
3. To provide an understanding of estimation in econometrics and its associated concepts.

Learning outcomes:

1. Students will gain an understanding of foundational knowledge of inference and estimation techniques.
2. Students will develop an intuition for the generalizability of different econometric models.
3. Students will understand the links between the theory for econometric models, the computations required for inference, and the application of the models to real examples.

Course content:

Revision of key statistical concepts: Descriptive statistics, Random variables, Joint distributions, conditional distribution and their properties, Moment generating functions, Normal distribution and other related distributions, Central Limit Theorem, Population, parameters and random sampling, sampling distributions, chi-square distribution, t distribution, F-distribution (6 lectures)

Estimation: Finite sample properties of estimators – unbiasedness and efficiency, Asymptotic or large sample properties of estimators – consistency, asymptotic normality (4 lectures)

General Approaches to parameter estimation: Method of moments, maximum likelihood estimation method, Least Squares method. Interval estimation and confidence intervals (3 lectures)

Linear Regression Model: multiple linear regression models, Gauss Markov Theorem, inference and hypothesis testing in the context of linear regression, multiple hypothesis testing, calculating marginal effects, mechanics and interpretation of ordinary least squares, Frisch-Waugh-Lovell Theorem, different econometric tests (LM test, LR test and Wald test) and their principles. (15 lectures)

Issues related to OLS: Heteroskedasticity and White test, Breusch-Pagan/Godfrey Test, Goldfeld-Quandt Test, Generalized Least Squares, Multicollinearity, Functional form misspecification and Ramsey RESET test, specification errors – errors of omission and errors of commission, restricted least squares, omitted variable bias, measurement errors and properties of OLS under measurement error, missing data, outliers and influential observations, sample-selection issues, Endogeneity and Instrumental variables. (15 lectures)

Reference Books:

1. "Introductory Econometrics: A Modern Approach" by Jeffrey Wooldridge, International Edition, Cengage

2. "Econometric Analysis of Cross-section and Panel Data" by Jeffrey Wooldridge, MIT Press, 2010

3. "Econometric Analysis" by William Greene, Pearson, 2018

4. Kennedy, P., *Guide to Econometrics* (6th ed.) MIT Press

5. Kmenta Jan, "Elements of Econometrics", Macmillan

Course Title	Introduction to Climate Change Economics	**Course No.**	LAL4850
Department/School	SoLA	**Structure**	
Offered for	UG	**Type**	
Pre-requisite		None	

Objectives:

- To equip students with the foundational knowledge of how economic tools and frameworks are applied to analyse climate change.
- To investigate the implications of climate change for economic development, with a focus on both global and regional disparities.
- To critically examine the design and effectiveness of economic instruments—such as taxes, subsidies, and market-based mechanisms—used in climate policy at the international, national, and local levels.

Course Outcome:

By the end of this course, students will be able to:

- Identify and explain key economic issues related to climate change, including externalities, public goods, and market failures.
- Apply economic reasoning to evaluate mitigation and adaptation strategies in response to climate risks.
- Demonstrate an understanding of major international climate agreements, negotiations, and institutional frameworks.
- Analyse the role of the circular economy and assess how recycling and resource efficiency contribute to climate resilience and sustainable development.

Contents:

- ***Science of Climate Change***: Economics, environmental economics, and ecological economics: why it is important to study environmental economics, Science of climate change: Process and projections, How and why the climate changed in the past, Climate change today, Climate vs weather, Types of climate and climate zones, Options for emission reductions, Cost of emissions reductions, Climate change and the environmental Kuznets curve. (8 lectures)
- ***Impacts of Climate Change***: Climate and economic development, How climate change affects people around the world, Implications of climate change for development, Impact of climate change on education, agriculture, human health, and ecosystem services, Costs of climate change in developed and underdeveloped countries, Economic modelling of climate change impacts. (8 lectures)
- ***Climate Change Adaptations and Beyond***: National and international adaptation policies, Adaptation vs mitigation, Carbon pricing and emission markets in practice, Accelerating technological innovation, Adaptation and development, The economics of adaptation, Adaptation in the developed and underdeveloped world, Role of adaptation in sustainable development, Optimal climate policy. (8 lectures)
- ***Climate Change Policies and Negotiations***: International environmental agreements, Cooperative and non-cooperative abatement, Free riding, Cartel formation, Multiple coalitions, IPCC, International climate policy, and negotiations over climate change policy. (8 lectures)
- ***Key economic issues related to climate change***: Externalities, public goods, and market failures, Role of the circular economy and how recycling and resource efficiency contribute to climate resilience and sustainable development, Recyclable resources: minerals, paper, bottles, and e-waste, managing municipal solid wastes (7 lectures)

Text Books:

- Tol, R. S. J. (2019). *Climate economics*. Edward Elgar Publishing.
- Houghton, J. T. (2004). *Global warming* (3rd ed.). Cambridge University Press.
- Stern, N. (2007). *The economics of climate change: The Stern review*. Cambridge University Press.

Reference Books:

- Hahn, R. W., & Ulph, A. (Eds.). (2012). *Climate change and common sense*. Oxford University Press.
- Harris, J. M., Roach, B., & Codur, A.-M. (2017). *The economics of global climate change*. Global Development and Environment Institute, Tufts University.
- Ackerman, F., & Stanton, E. (2013). *Climate economics*. Routledge.

Online Reading Materials

- Economics and policies of climate change. Coursera. https://www.coursera.org/learn/economics-and-policies-of-climate-change
- The complete starter guide to climate change. Udemy. https://www.udemy.com/course/climate-change
- Climate change 2022: Impacts, adaptation and vulnerability. IPCC. https://www.ipcc.ch/report/sixth-assessment-report-working-group-ii/
- Climate change 2022: Mitigation of climate change. IPCC. https://www.ipcc.ch/report/sixth-assessment-report-working-group-3/

Title	**Health Economics**	Number	LAL4070
Department	Humanities and Social Sciences	L-T-P [C]	3-0-0[3]
Offered for	UG	Type	Elective
Prerequisite	None		

Objective

1. To be able to evaluate the health sector in India using economic tools.
2. To have a global perspective and a broad understanding of institutional changes currently in the Indian healthcare system.

Learning Outcomes

1. Understand microeconomic concepts and their application for the health sector.
2. Understand the role of economic incentives, competition and regulation on the functioning of the health care system.
3. Understand the health care insurance sector and financing of health care.
4. Understand how different healthcare systems function around the world.

Contents

Introduction to health economics*:* An introduction to health, health care, health system and health economics (*5 lectures*).

Microeconomic and statistical concepts*:* Demand, supply, elasticity, production, market structures, economic efficiency & cost-benefit analysis, economic evaluation, and valuation of human life, hypothesis testing, regression analysis, statistical inference (*12 lectures*).

Supply and Demand for health*:* Health production function and cost of health care, health demand, human capital model and the determinants of demand for health, demand for and supply of health insurance, uncertainty, adverse selection and moral hazard in the health insurance market (*10 lectures*).

Healthcare Industry*:* Market Structure of health care industry, types and characteristics of various health care providers, India's healthcare system (*5 lectures*).

Government and Health care systems*:* Healthcare financing, rationale for government intervention in health care market and the nature of intervention by government in health care sector, Comparison of healthcare systems in different countries, Universal Health Coverage in India (*10 lectures*).

Reference Books

1. Introduction to health economics. Guinness, & Wiseman. McGraw-Hill Education (UK), 2011.
2. McPake, B., Normand, C., Smith, S., & Nolan, A. (2020). *Health economics: an international perspective*. Routledge.
3. *The Economics of Health and Health Care*, Sherman Folland, Pearson, 7/e, 2012
4. *Health Economics for Developing Countries: A Practical Guide*, Witter S, T Ensor, M Jowett, and R Thompson, McMillan Education Ltd, 2000

Electronic Resource

http://www.healtheconomics.com/resources/

Title	**Decision Sciences Applications**	Number	**AIL6110**
Department/School	AIDE (CMCE)	L-T-P [C]	3-0-0
Offered for	B. Tech./P.G.	Type	Elective
Prerequisite	None		

Objectives:

1. To introduce students to decision sciences and operations research.
2. To provide practical motivation for linear programming and its applications.
3. To understand the solution methods and its implementation.

Learning outcomes:

1. Formulate different types of business situations using linear programs
2. Implement linear programs using different tools and packages
3. Summarize duality theory and its economic interpretation
4. Solve special types of linear programs and their applications

Course Content:

Linear Programming: Formulation and Examples, Intuition, Modeling Business Situations, Geometric Motivation and Analysis, Implementation using Different Tools and Packages, Solution Methods (8 lectures). [RB1, RB3, OR]

Economic Applications: Game Theoretic Applications, Queueing and Inventory Models, Data Envelopment Analysis, Network Problems, Other Applications (18 lectures). [RB1, RB4]

Duality for Policy Making: Introduction, Formulation, Economic Interpretation, Duality Applications and Prescriptive Analytics (16 lectures). [RB1, RB2, RB4]

Textbook:

1. Srinivasan, G. (2017). Operations Research: Principles and Applications. PHI Learning Pvt. Ltd.

Reference books:

1. Beilby, M. H. (2014). *Economics and Operational Research*. Academic Press.
2. Solow, D. (2014). Linear programming: An introduction to finite improvement algorithms. Courier Corporation.
3. Thie, P. R., & Keough, G. E. (2011). An introduction to linear programming and game theory. John Wiley & Sons.

Online Resources:

Srinivasan, G. NPTEL course on Operations Research - https://archive.nptel.ac.in/courses/112/106/112106134/

Title	**Operations Research for Economics**	Number	**AIL7160**
Department/School	AIDE (CMCE)	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech./P.G./Ph.D.	Type	Elective
Prerequisite	None		

Objectives:

1. To introduce students to operations research and its role in economics.

2. To provide conceptual understanding of operations research as a tool to solve problems in economics, business and policy making.

Learning outcomes:

1. Formulate economic and business problems using operations research tools

2. Implement operations research tools to solve various economics, business, and policy problems

Course content:

Business formulation: Introduction to Operations Research and Economics, Concepts and examples, Case-based applications, Solution methods (12 lectures) [RB1, RB2, RB5]

Applications to game theory: Computing equilibria, Matrix games - Computing mixed strategy Nash equilibrium, Cooperative games - Computing Core using linear program, Queueing Models, Strategic Queueing (10 lectures) [RB2, RB3, RB6]

Supply chain contracts: Inventory models, Newsvendor model, Supply chain – structure and concepts, Wholesale price contract, Buyback contract and Revenue sharing contract (6 lectures) [RB4]

Policy applications: Comparing the performance of decision making units, Prescription for inefficient units using data envelopment analysis, Other applications (14 lectures) [RB1]

Textbook:

1. Narahari, Y. (2014). *Game theory and mechanism design* (Vol. 4). World Scientific.
2. Srinivasan, G. (2017). Operations Research: Principles and Applications. PHI Learning Pvt. Ltd.

Reference books:

1. Vohra, R. V. (2004). Advanced mathematical economics. Routledge.
2. Chopra, S. & Meindl, P. (2015). Supply Chain Management: Strategy, Planning and Operation. Pearson Education.
3. Beilby, M. H. (2014). *Economics and Operational Research*. Academic Press.
4. Hassin, R., & Haviv, M. (2003). To queue or not to queue: Equilibrium behavior in queueing systems (Vol. 59). Springer Science & Business Media.

Title	**Fundamentals of Economics**	Number	**AIL4070**
Department/School	AIDE (CMCE)	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech	Type	Core
Prerequisite	None		

Objectives:

1. To introduce the concepts as – learning and unlearning to think like an economist, the fundamentals of economics, and macroeconomics in closed and open economy.
2. To explore how the state can facilitate the functioning of the market economy, development, globalization, and how ideas and machines are useful for economic development and growth.

Learning outcomes:

1. Students will be familiarized with how the basic economic thinking unfolds, how economic agents decide, interact, and how the state can facilitate the functioning of the market economy.
2. Students will understand the functioning of the economics of institutions, development, globalization, and the importance of technology for economic development and growth.

Course Contents:

1. Learning and Unlearning to Think Like an Economist: Principles of Economics**,** Thinking like an Economist**,** Strategic Thinking, Prisoner's Dilemma and Applications. (L11)

2. Unfolding the Subject Matter of Economics: Market Forces of Demand and Supply, Elasticity and its Applications, Supply Demand and Government Policies. (L4)

3. Money in the Modern Economy: History and Concept of money; monetary aggregates; Theories of Demand for money; money supply and credit creation. (L3)

4. Unfolding the Subject Matter of Macroeconomics: Macroeconomic Concerns, The Components and Measurement of Macroeconomic Data, Circular flow, Keynesian Income Determination, Multipliers, Open Economy and Balance of Payments, Exchange Rates. (L9)

5. Finance and Macroeconomy: The Financial System, Economic Growth, and Financial Crises. (L3)

6. Institutions, Development, Globalization, and Inequality: The evolution of democracy and state**,** Conditions for Modern Democracy and Citizenship, Globalization. (L8)

7. Economics of Ideas and Technology: Economics of Ideas, Economics and AI. **(4)**.

Reference Books:

1. Mankiw, N. G. (2021). *Principles of economics*. Cengage Learning.
2. Karl, E., CASE, F., OSTER, R. C., & Sharon, E. (2017). *Principles of Economics*. Pearson.
3. Jones, C., & Vollrath, D. (2013). Introduction to Economic Growth New York, US: WW Norton & Company.